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Objective: Add roadNet-CA support and run all 6 variants on it for comparison.

Attempts made: roadNet-CA is formatted differently from LiveJournal. No weight column and each line represents an undirected edge so both directions need to go into the CSR. Wrote a second loader function in graph.cpp for this. For weights I just picked a random value per edge in [1, 400] using mt19937 seeded at 42 and assigned the same weight to both directions so SSSP stays consistent.

Got the file with wget and it downloaded fine. Added it to .gitignore since its too big to commit. Then added a section at the bottom of main.cpp to run all 6 variants on the roadNet graph.

Solutions/Workarounds: Ran into a problem where graph.cpp only saved partway through. The new loader function was there but cut off after the first few lines so it wouldn't compile. Ended up overwriting the whole file from scratch using cat with a heredoc which got everything in cleanly. Probably happened because I was pasting a large block into the editor.

Commits: exp: add roadNet-CA graph loading and timing; exp: run all variants on roadNet-CA

Experiments/Analysis: roadNet-CA has 1,971,281 vertices and around 5.5M edges after inserting both directions. Results with 4 threads:

roadNet BFS dense atomic (4t): 3.820 sec roadNet BFS sparse atomic (4t): 0.210 sec roadNet SSSP dense atomic (4t): 8.354 sec roadNet SSSP sparse atomic (4t): 3.407 sec roadNet CC dense atomic (4t): 10.327 sec roadNet CC sparse atomic (4t): 4.797 sec

Sparse wins by a lot on roadNet, way more than on LiveJournal. BFS sparse is about 18x faster than BFS dense here. On LiveJournal they were almost the same. The difference comes down to graph structure. LiveJournal is a social network so it has a small diameter and high degree hubs, meaning BFS reaches most of the graph in just a few rounds and the frontier blows up fast. roadNet is a road network so degree is low and diameter is huge, the frontier stays small for hundreds of rounds. That's where the sparse worklist actually saves time since you're only touching active vertices instead of scanning all 1.9M every round.

CC on roadNet took 549 rounds for sparse vs just 7 on LiveJournal. The road network propagates labels slowly because of the grid-like topology so CC has to grind through many rounds before converging.

Learning outcome: The performance gap between dense and sparse depends heavily on the graph. On LiveJournal the frontier gets so large so fast that the sparse overhead doesn't help much. On roadNet the frontier is small for most of the run so sparse is the clear winner. This is basically the motivation for the direction-optimizing bonus where you switch between dense and sparse each round based on how big the frontier is.